Chapter 3: Graphical Linkage Synthesis

- Qualitative, quantitative, and dimensional synthesis
- Function, path, and motion generation
- Limiting conditions: toggle positions, transmission angle
- Two-position synthesis (rocker and coupler output)
- Three-position synthesis (with specified and free pivots)
- Quick-return mechanisms (fourbar and sixbar)
- Coupler curves and their properties
- Cognates and Roberts' theorem
- Straight-line mechanisms (Watt, Chebyschev, Hoeken, etc.)
- Dwell mechanisms

Related Reading: Norton, Ch. 3, pp. 98–177.

Types of Synthesis

Qualitative synthesis: Creation of potential solutions *in the absence of* a well-defined algorithm. More unknowns than equations $\rightarrow$ requires engineering judgment.

Quantitative (analytical) synthesis: Generation of solutions using a defined algorithm. Equations exist; a numerical answer is obtained. Quality still requires judgment.

Type synthesis: Determine the *proper type* of mechanism best suited to the problem (linkage vs. cam vs. gear vs. cylinder, etc.).

Dimensional synthesis: Determine the *proportions (link lengths)* necessary to accomplish the desired motions. This chapter: **graphical** methods. Chapter 5: analytical methods.

[Norton, pp. 98–100]

Type Synthesis: Choosing the Right Mechanism

Example: Track a part on a conveyor belt and spray a chemical coating. Requirements: straight-line motion, high speed, accuracy, repeatability, low cost.

Possible solutions:

- A straight-line linkage
- A cam and follower
- An air cylinder / hydraulic cylinder
- A robot / solenoid

Each has trade-offs in cost, precision, reliability, and complexity.

Design is essentially an exercise in trade-offs. Seldom will there be one clear, obvious solution. Remember: *an engineer can do, with one dollar, what any fool can do for ten dollars.*

[Norton, pp. 99–100]

Function, Path, and Motion Generation

Function generation: Correlation of an *input motion with an output motion* in a mechanism. The mechanism is a "black box" that delivers a predictable output for a known input.

Path generation: Control of a *point* in the plane such that it follows a prescribed path. Orientation of the link is *not* controlled. (Coupler curves.)

Motion generation: Control of a *line* (or body) in the plane such that it assumes prescribed sequential positions. Both position *and* orientation are controlled. This is the most general case; path generation is a subset.

Planar vs. spatial: We focus on planar (2-D) mechanisms. Many 3-D devices are constructed from **duplicate planar linkages** in parallel planes.

[Norton, pp. 100–101]

Limiting Conditions: Toggle Positions

After synthesizing a linkage, you **must** check for limiting conditions.

Toggle positions: Positions at the limits of a fourbar linkage's motion, determined by the *colinearity of two of the moving links*.

- Non-Grashof triple-rocker: has 4 toggle positions.
- Grashof double-rocker: has 2 toggle positions per driven link.

At a toggle, the linkage cannot be driven through by force applied to the input link — it must be pushed "over center" by a different link or by momentum (e.g., a flywheel).

After synthesizing a multi-position linkage, you **must** check for toggle positions *between* your design positions. A CAE tool (e.g., LINKAGES) will check this automatically.

Toggles can be useful: self-locking feature (e.g., pickup truck tailgate, ironing board legs).

[Norton, pp. 102–103]

Limiting Conditions: Transmission Angle

The **transmission angle** μ is the angle between the *output link* and the *coupler*. It is a measure of the quality of force and velocity transmission at the joint.

The force $\mathbf{F}_{34}$ transmitted from coupler to output decomposes into:

- **Tangential component** $F_{34}^t = F_{34} \sin \mu \rightarrow$ creates output torque.
- **Radial component** $F_{34}^r = F_{34} \cos \mu \rightarrow$ only creates bearing friction.

Optimum: $\mu = 90^\circ$ (all force creates torque).
Minimum acceptable: $\mu \geq 40^\circ$ for most designs. Below this, force transmission is poor and the linkage may stall or have jerky motion.

The transmission angle varies continuously as the linkage moves. You must check μ_{min} over the full range of motion.

[Norton, pp. 103–105]

Two-Position Synthesis: Overview

Position synthesis: Determination of the link lengths necessary to accomplish the desired motions. Graphical methods work well for **up to three** design positions. Beyond three $\rightarrow$ analytical/computer methods (Chapter 5).

Two-position synthesis has two subcategories:

Output type	**Problem**	**Generation**
Rocker output	Pure rotation of a link	Function gen.
Coupler output	Complex motion of a line	Motion gen.

Tools needed: Only a compass, protractor, and straightedge (Euclidean geometry). The key construction is **perpendicular bisection** of line segments.

[Norton, pp. 105]

Two-Position Synthesis: Rocker Output (Example 3-1)

Problem: Design a Grashof crank-rocker with 45° of rocker rotation and equal time forward and back.

Method:

1. Draw output link O_4B in both extreme positions (B_1, B_2) subtending θ_4.
2. Draw chord B_1B_2 and extend it.
3. Select a convenient point O_2 on line B_1B_2 extended.
4. Bisect segment B_1B_2 and draw a circle of that radius about O_2.
5. The two intersections with the extended chord give A_1 and A_2.
6. Measure coupler (A_1B_1), crank (O_2A_1), rocker (O_4B_1), ground (O_2O_4).
7. Check Grashof condition. If non-Grashof, move O_2 farther from O_4 and repeat.

Key insight: We start from the *output end* (the only defined aspect) and work backward. Many "free choices" exist — these are qualitative design decisions that affect quality.

[Norton, pp. 105–107]

Two-Position Synthesis: Coupler Output (Example 3-3)

Problem: Move link CD from position C_1D_1 to C_2D_2 (motion generation).

Method:

1. Draw C_1D_1 and C_2D_2 in the plane.
2. Draw construction lines C_1C_2 and D_1D_2.
3. Bisect C_1C_2 and D_1D_2; extend perpendicular bisectors.
4. Select any convenient point on each bisector as the fixed pivots O_2 and O_4.
5. Connect: O_2C_1 = link 2, O_4D_1 = link 4, C_1D_1 = link 3 (coupler), O_2O_4 = link 1 (ground).
6. Check Grashof condition and transmission angles.

The coupler output solution is often a **non-Grashof triple-rocker**. To drive it, add a **dyad** (twobar chain) to create a Watt sixbar mechanism with a motor-driven crank.

[Norton, pp. 107–110]

The Rotopole and Driver Dyads

Rocker output with complex displacement (Example 3-2): To move link CD through two positions using a *rocker*, find the **rotopole** — the intersection of the perpendicular bisectors of C_1C_2 and D_1D_2. This reduces the problem to Example 3-1.

Adding a driver dyad (Example 3-4):

- When the fourbar is a double- or triple-rocker, it cannot be driven by a simple crank.
- Solution: add a **dyad** (2-link chain) as a driver stage $\rightarrow$ creates a **Watt sixbar**.
- The driver subchain must be a **Grashof crank-rocker**.
- There are an *infinity* of driver dyads possible; the motor pivot O_6 can be placed at any convenient location.

[Norton, pp. 107–110]

Three-Position Synthesis

Three-position synthesis defines 3 positions of a line in the plane and creates a fourbar to move it through all three. This is a **motion generation** problem.

Three variants:

Variant	**What you specify**
Specified moving pivots	3 positions of coupler $CD \rightarrow$ find fixed pivots
Alternate moving pivots	3 positions of CD; use different attachment points EF
Specified fixed pivots	Fixed pivots O_2, $O_4 \rightarrow$ find moving pivots via **inversion**

Method (specified moving pivots): Same perpendicular bisector technique as two-position, but applied to three positions ($C_1C_2C_3$ and $D_1D_2D_3$). The bisectors of C_1C_2 and C_2C_3 intersect at O_2; similarly for D points.

Must always check for toggles **between** design positions!

[Norton, pp. 111–114]

Three-Position Synthesis with Specified Fixed Pivots

When fixed pivot locations are constrained by the package, use **inversion**:

Key idea: Instead of the coupler moving relative to the ground, pretend the *ground plane moves* relative to the coupler. This swaps the roles of coupler and ground.

Method (Examples 3-7 and 3-8):

1. Draw all 3 coupler positions (C_1D_1, C_2D_2, C_3D_3) and the ground link O_2O_4.
2. Transfer the ground link position to each coupler position using arc constructions $\rightarrow$ find the 3 "inverted" ground positions.
3. These 3 inverted positions (E_1F_1, E_2F_2, E_3F_3) define a standard 3-position synthesis problem.
4. Solve for "fixed" pivots G and H (which are really moving pivots on the coupler).
5. **Reinvert**: the moving pivots become points on link 3, and O_2O_4 resumes its role as ground.

[Norton, pp. 114–118]

Beyond Three Positions

Four-position synthesis does not lend itself to manual graphical solutions. It requires:

- A set of simultaneous vector equations representing all 4 positions.
- Free choices of some variables, then numerical solution.
- Computer programs: LINCAGES (Erdman), KINSYN (Kaufman), LINKAGES (Norton).

See Chapter 5 for the analytical approach.

General principle: the more constraints imposed, the harder the synthesis. Each additional position dramatically increases difficulty. Graphical methods are practical for ≤ 3 positions.

[Norton, pp. 119]

Quick-Return Mechanisms

Many machines need different speeds for "forward" (working) and "return" strokes.

Time ratio T_R: the ratio of time for the forward stroke to the return stroke.

$$T_R = \frac{\alpha}{\beta}, \qquad \alpha + \beta = 360^\circ \qquad \therefore \quad T_R = \frac{\alpha}{360^\circ - \alpha} \tag{3.1}$$

Construction angle:

$$\delta = |180^\circ - \alpha| = |180^\circ - \beta| \tag{3.2}$$

A **non-quick-return** linkage has O_2 on the chord B_1B_2 extended $\rightarrow$ $\alpha = \beta = 180^\circ \rightarrow T_R = 1$.
A **quick-return** has O_2 off the chord $\rightarrow \alpha \neq \beta \rightarrow T_R \neq 1$.

[Norton, pp. 119–120]

Fourbar and Sixbar Quick-Return Designs

Fourbar quick-return (Example 3-9):

- Crank-rocker with specified T_R and rocker angle θ_4.
- Works well for T_R down to about 1:1.5.
- Below that, transmission angles become poor.

Sixbar drag-link quick-return (Example 3-10):

- For larger time ratios (up to $\sim$ 1:2).
- First stage: fourbar **drag link** (double-crank) with the desired T_R.
- Second stage: **dyad** driven by the "dragged" crank, producing either rocker or slider output.

Crank-shaper quick-return (Fig. 3-14):

- Inversion #2 of the crank-slider.
- Capable of very large time ratios.
- Used in metal shaper machines for slow cutting / fast return.

[Norton, pp. 119–124]

Coupler Curves

The **coupler** is the most interesting link in any linkage — it is in complex motion and any point on it traces a **coupler curve**. The fourbar coupler curve is a 6th-degree algebraic curve.

$$m = 2 \cdot 3^{(n/2-1)} \tag{3.3}$$

gives the max degree m for a mechanism with n links connected by revolute joints: fourbar $\rightarrow$ degree 6, sixbar $\rightarrow$ 18, eightbar $\rightarrow$ 54.

Coupler curves can approximate:

- **Straight lines** (approximate or exact)
- **Circular arcs** with remote centers
- **Complex paths** with cusps, crunodes, and loops

Cusp: a sharp point where velocity is instantaneously zero $\rightarrow$ useful for transporting, stamping, feeding operations.
Crunode: a point where the curve crosses itself, creating multiple loops.

[Norton, pp. 124–126]

Coupler Curve Shapes and the H&N Atlas

"Cursory catalog" of coupler curve shapes: pseudo-ellipse, kidney bean, banana, crescent, single straight, double straight, teardrop, scimitar, umbrella, triple cusp, figure eight, triple loop.

The Hrones and Nelson (H&N) Atlas:

- $\sim$7000 coupler curves for Grashof crank-rocker linkages.
- Organized by link ratios (unit-length crank), 50 coupler points per linkage, 10 linkages per page.
- Each dashed station $= 5^\circ$ of crank rotation $\rightarrow$ shows velocity variation.
- Invaluable starting point for path generation problems.

Symmetrical fourbar coupler curves: When $L_3/L_2 = L_4/L_2 = BP/L_2$ (equation 3.4: $AB = O_4B = BP$), only 3 parameters define the geometry instead of 5, greatly simplifying the design space.

[Norton, pp. 125–132]

Practical Application: Film Advance Mechanism

Movie camera film advance (Fig. 3-18): A Grashof crank-rocker with a hook on the coupler at point C:

1. At point F_1: the hook enters a film sprocket hole (motion nearly perpendicular to film).
2. The coupler then pulls the film downward along an approximately straight line.
3. At point F_2: a **cusp** on the coupler curve causes the hook to decelerate, reverse, and smoothly exit the sprocket hole.
4. The remaining curve is "wasting time" — film advances under inertia while the shutter closes and reopens.

This example illustrates the power of coupler curves: a simple 4-link mechanism, with only 4 pins and no sliding joints, produces a complex and precisely timed motion from a single constant-speed motor.

[Norton, pp. 128]

Cognates and Roberts' Theorem

Roberts' theorem (1875): Any coupler curve of a fourbar linkage can be generated by **three different fourbar linkages** (called **cognates**). These three linkages share the same coupler curve but have *different fixed pivot locations*.

This is extremely useful when a good coupler curve solution has fixed pivots in an inconvenient location.

Cayley diagram construction:

1. Start with the original fourbar and its coupler point P.
2. Align links 2 and 4 with the coupler to form a reference triangle.
3. Construct lines parallel to all sides of the original fourbar.
4. The intersections define the two cognate linkages.
5. All three share similar triangles: $\triangle A_1 B_1 P \sim \triangle A_2 P B_2 \sim \triangle P A_3 B_3$.

[Norton, pp. 134–137]

Cognates: Properties and Applications

Properties of cognates (Nolle/Luck):

- If the original is a Grashof **crank-rocker**, one cognate is also a crank-rocker (same μ_{min}); the other is a Grashof double-rocker.
- If the original is a non-Grashof linkage, the cognates may have more favorable characteristics.

Practical use: Parallel motion from cognates

- Combine cognate #1 with cognate #3 (sharing a common coupler point P).
- Merge overlapping links $\rightarrow$ results in a **Watt I sixbar** that produces **parallel motion** (curvilinear translation) of a coupler point.
- Useful when you need a point to follow a path while maintaining orientation.

Geared fivebar cognates: A fourbar cognate can also be converted to a geared fivebar mechanism, providing yet another configuration option.

[Norton, pp. 137–140]

Straight-Line Mechanisms

One of the oldest problems in kinematics: generate straight-line motion from rotary input. Two categories:

Approximate straight-line linkages:

- **Watt's linkage** (1784): the first straight-line mechanism, invented for the steam engine. Uses the coupler of a non-Grashof double-rocker. Approximate.
- **Chebyschev linkage**: a Grashof double-crank (drag link). Link ratios $L_1 : L_2 : L_3 = 1 : 2.5 : 2.5$, ground $= 2$.
- **Hoeken linkage**: a Grashof **crank-rocker** (practical advantage: can be continuously driven). Also provides nearly **constant velocity** along the center of the straight portion.

Exact straight-line linkages:

- **Peaucellier** (1864): 8 links — the first exact straight-line linkage.
- **Watt-Evans**: 6 links, exact but limited range.
- Generally more complex $\rightarrow$ more joints $\rightarrow$ more error in practice.

[Norton, pp. 141–148]

The Hoeken Linkage: Optimized Straight-Line Design

The Hoeken linkage is the most practical straight-line fourbar because:

- It is a **Grashof crank-rocker** (can be motor-driven continuously).
- It provides **nearly constant velocity** along the straight portion.
- Table 3-1 gives optimized link ratios for various crank-angle ranges.

Two optimization criteria (you cannot have both simultaneously):

1. **Minimum straightness error:** minimize deviation from a true straight line.
2. **Minimum velocity error:** minimize deviation from constant velocity along the line.

Table 3-1 provides link ratios for each criterion across crank-angle ranges from 20° to 180°.

The position and velocity structural errors are typically $< 1\%$ for moderate crank-angle ranges ($\leq 100^\circ$).

[Norton, pp. 143–148; Table 3-1]

Dwell Mechanisms

A **dwell** is a period during which the output link remains stationary while the input continues to move. Dwell mechanisms are needed for indexing, assembly, and intermittent-motion applications.

Single-dwell sixbar (Fig. 3-31):

1. Find a fourbar coupler curve with a **pseudo-circular-arc** segment over the desired portion of crank rotation.
2. Add a dyad whose fixed pivot is at the approximate center of that arc.
3. During the arc portion: the output link dwells (the coupler point moves on a circle centered at the output pivot).
4. During the rest: the coupler point drives the output through its working stroke.

The dwell will have some **"jitter"** because the pseudo-arc is only an approximation to a true circle.

[Norton, pp. 148–152]

Double-Dwell Mechanisms

Double-dwell sixbar (Fig. 3-32):

- Requires a coupler curve with **two** approximately straight segments.
- Add a dyad with a **slider in a slot** (link 6) whose pivot is tangent to both straight portions.
- During each straight portion: the slider moves along link 6 without rotating it $\rightarrow$ output dwells.
- Between the straight portions: the coupler drives the output through its stroke.

Limitations of linkage dwells:

- Always have some jitter (not exact dwells).
- Tend to be physically large relative to output motion.
- Output acceleration can be very high.
- Cams give *exact* dwells but have their own limitations.

See Chapter 8 for cam-based dwell designs.

[Norton, pp. 151–153]

Other Useful Linkages

Constant-velocity piston motion (Fig. 3-34):

- Watt II sixbar linkage designed so that the slider moves at nearly constant velocity over a portion of the stroke.
- Useful for applications requiring uniform coating or feed rates.

Large angular excursion rocker (Fig. 3-35):

- Stephenson III sixbar that gives 180° oscillation of the output rocker from a fully rotating crank input.
- Useful for large-displacement reciprocating mechanisms.

Remote center circular motion (Fig. 3-36):

- Eightbar linkage providing -360° oscillatory rotation of output link.
- Emulates a large-radius circular arc without a physical pivot at the center.

[Norton, pp. 153–157]

Chapter 3: Summary of Key Concepts

1. **Three types of synthesis:** qualitative, quantitative, type/dimensional.
2. **Three types of generation:** function, path, and motion.
3. **Limiting conditions:** always check toggle positions and transmission angle ($\mu_{\text{min}} \geq 40°$).
4. **Two-position synthesis:** perpendicular bisector method. Rocker output or coupler output. Add a driver dyad if needed.
5. **Three-position synthesis:** extension of two-position method. Use **inversion** to specify fixed pivot locations.
6. **Quick-return:** time ratio $T_R = \alpha/\beta$; fourbar (up to 1:1.5), sixbar drag-link (up to 1:2), crank-shaper (larger ratios).
7. **Coupler curves:** 6th-degree curves (fourbar); cusps and crunodes; H&N atlas is essential.
8. **Roberts' theorem:** three cognate fourbars share the same coupler curve.
9. **Straight-line linkages:** Watt, Chebyschev, Hoeken (practical best), Peaucellier (exact).
10. **Dwell mechanisms:** sixbar linkages using pseudo-arc or straight-line coupler curve segments.

Key Equations to Remember

Time ratio	$T_R = \frac{\alpha}{\beta} = \frac{\alpha}{360^\circ - \alpha}$
Construction angle	$\delta = \lvert 180^\circ - \alpha \rvert = \lvert 180^\circ - \beta \rvert$
Max coupler curve degree	$m = 2 \cdot 3^{(n/2-1)}$
Symmetrical fourbar criterion	$AB = O_4B = BP$
Transmission angle rule	$\mu_{\text{min}} \geq 40^\circ$ (optimum: 90°)
Roberts' theorem	3 cognate fourbars share 1 coupler curve

Key Definitions to Remember

Qualitative synthesis	Creative, no algorithm; judgment-based
Quantitative synthesis	Algorithm-based; numerical answers
Function generation	Correlate input motion $\rightarrow$ output motion
Path generation	Control path of a point (not orientation)
Motion generation	Control position *and* orientation of a body
Toggle position	Limit of motion; two links become colinear
Transmission angle	Angle between coupler and output link
Coupler curve	Path traced by any point on the coupler
Cusp	Sharp point; instantaneous zero velocity
Crunode	Self-intersection of coupler curve
Cognate	Different fourbar sharing the same coupler curve
Rotopole	Intersection of perpendicular bisectors
Dyad	Twobar chain added as a driver stage
Time ratio	α/β; measure of quick-return

Self-Check Questions

Test your understanding of Chapter 3:

1. What is the difference between function, path, and motion generation?
2. What is the transmission angle and what is its minimum acceptable value? Why?
3. Describe the perpendicular-bisector method for two-position synthesis with rocker output.
4. What is a toggle position? Why must you check for them between design positions?
5. How does the "inversion" technique allow you to specify fixed pivot locations in three-position synthesis?
6. Define the time ratio. Derive T_R from α and β.
7. Name three types of quick-return mechanisms and their typical time-ratio ranges.
8. What is a coupler curve? What degree is it for a fourbar linkage?
9. State Roberts' theorem. How is it useful in practice?
10. Compare the Watt, Chebyschev, and Hoeken straight-line linkages. Which is most practical and why?
11. What is the difference between a single-dwell and double-dwell sixbar mechanism?
12. Why is the Hoeken linkage preferred over exact straight-line